IMAG/e

TU/e Technische Universiteit Eindhoven University of Technology

Generative Models part II

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Topics:

- Addressing VAE limitations
- GANs
- Conditional generation
- Image-to-image translation
- Limitations and extensions to GANs
- Fréchet inception distance
- Diffusion models

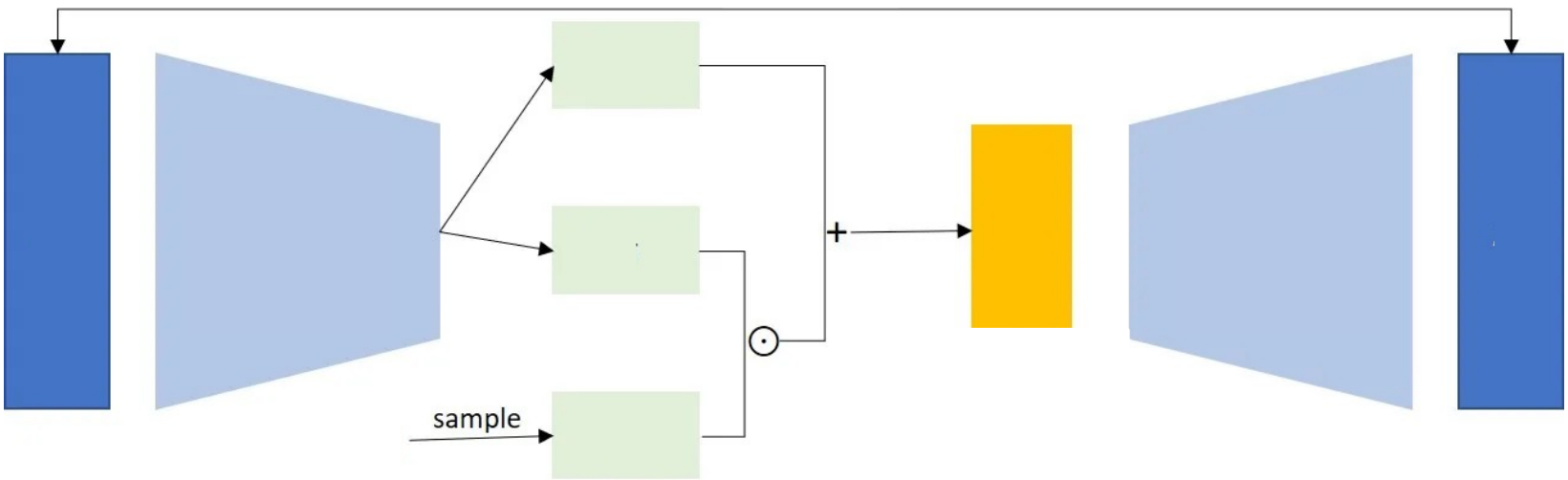
Intended learning outcomes

The student can:

- Discuss the intuition behind generative adversarial networks (GANs)
- Understand the benefits of GANs in the context of the limitations of VAEs
- Formulate the GAN training process and loss function
- Classify applications of GANs
- Give examples of issues with GANs
- Motivate solutions to these issues
- Relate diffusion models to VAE

Recap VAEs

Minimize 1:



Minimize 2:

Recap VAEs

- A VAE is a latent variable model
  - Sample $\mathrm{z} \sim p(\mathrm{z})$
  - Generate $\mathrm{x} \sim p(\mathrm{x}|\mathrm{z})$

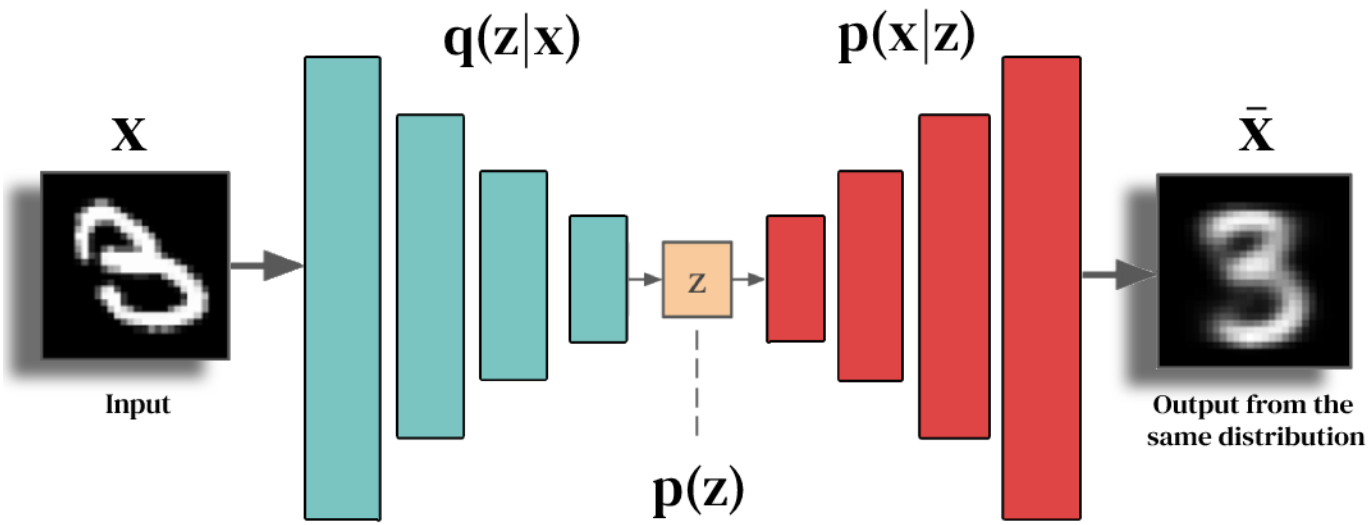
- A VAE is an *explicit* likelihood model
  - It tries to explicitly compute (approximate) the density $p(\mathrm{x})$
  - But we need to do a lot of work to get around the lack of an analytical solution and the curse of dimensionality

Limitation:

- Intractable likelihood - > Maximise ELBO rather than likelihood directly

Limitations:

- VAEs may not generate samples that are as high-quality or realistic, especially for images.



- Classes may overlap in latent space. (Latent space is entangled).
- Some points in latent space may generate samples that do not make sense (e.g. a combination of the shapes for two different numbers in the MNIST example)

Generative adversarial networks (GANs)

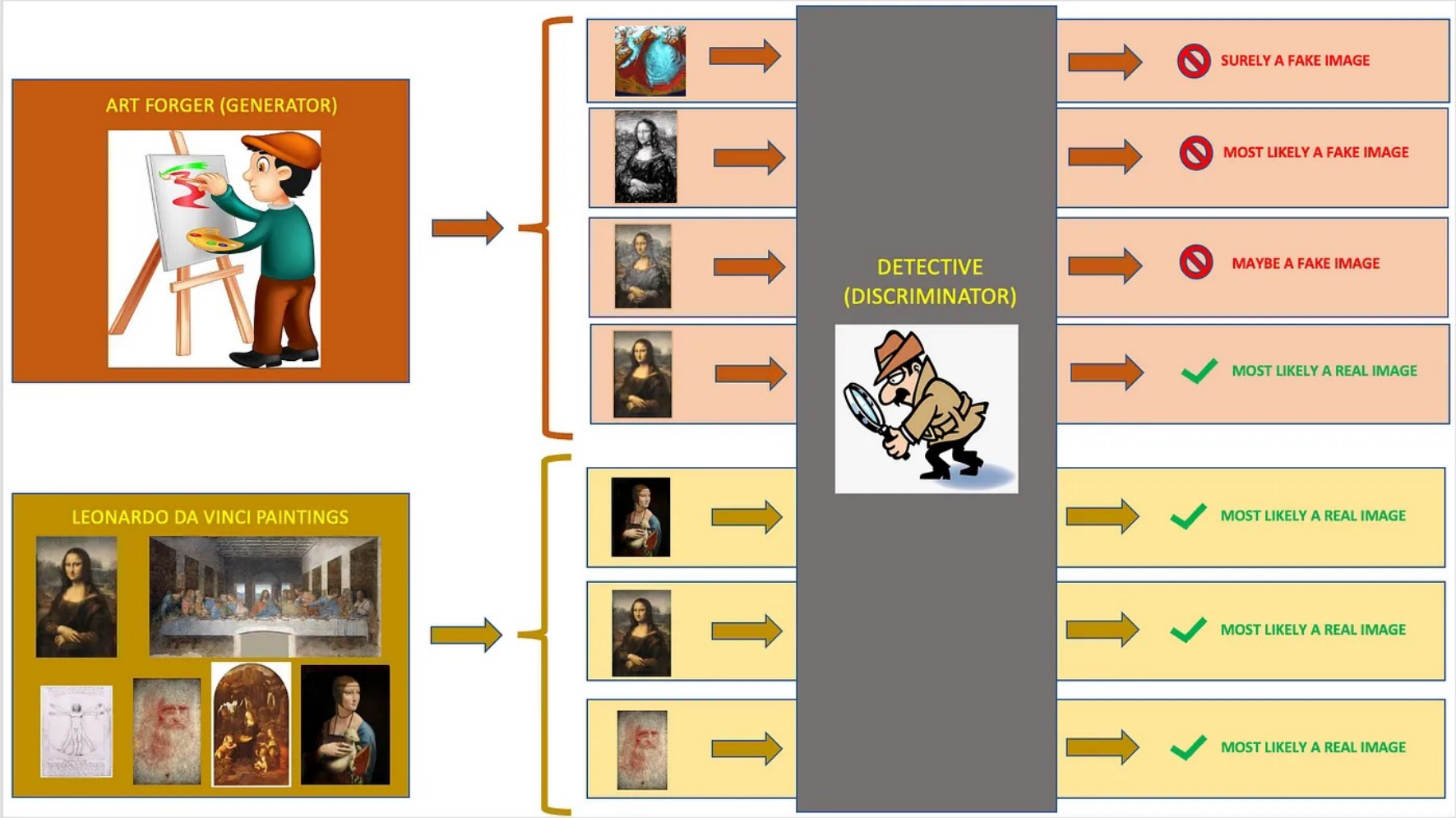
Previously:

- We used the log-likelihood to describe the differences between generative samples and the training distribution during training.
- As well as the practical difficulties, it is unclear if better likelihood numbers correspond to better samples.

Now:

- Is there another way?
- I.e. instead of using the log-likelihood, is there another way to calculated the differences between real and generate data?
- Can we learn it?

GAN intuition:

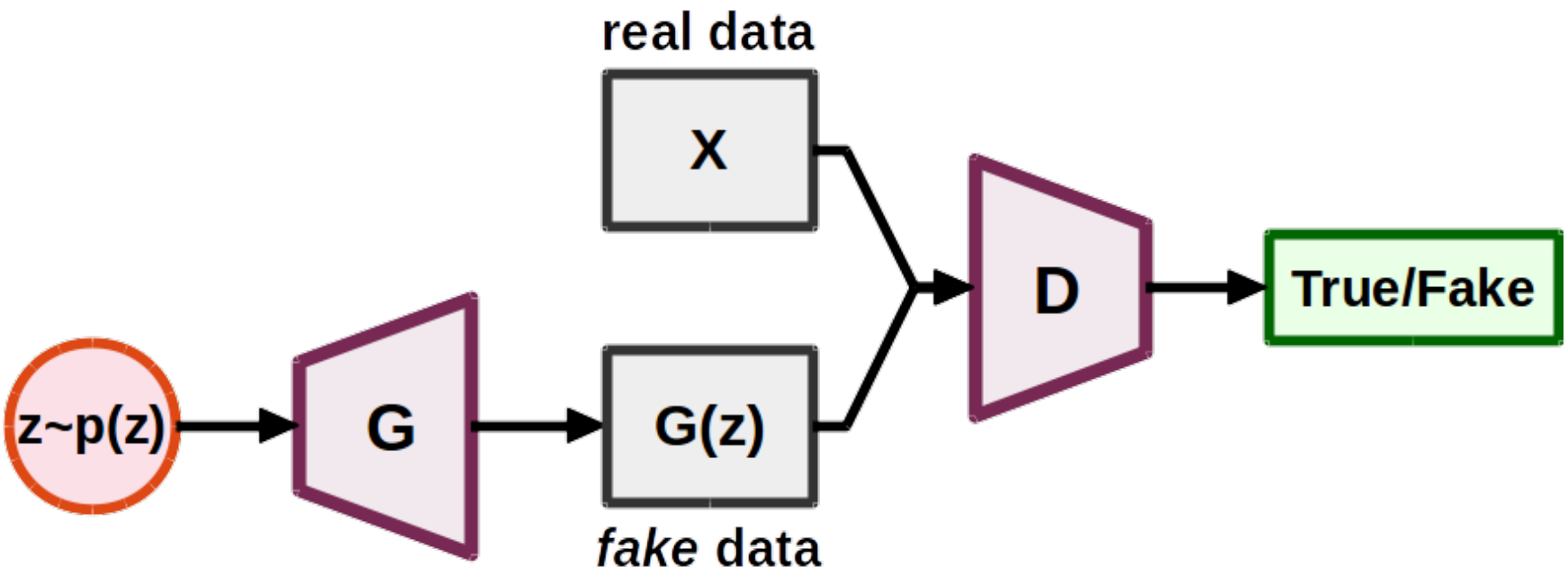


Generative adversarial networks (GANs)

Main idea:

- Train a classifier to distinguish samples as either real or generated
- Train the generator to create realistic samples that fool the classifier

GAN:



Algorithm:

- Generator - $G_\beta: Z \rightarrow X$.
- Discriminator - $D_\alpha: X \rightarrow [0, 1]$.
- Discriminator tries to maximize $\log D_\alpha(x) + \log 1 - D_\alpha\big(G_\beta(z)\big)$ with respect to α.
- Generator tries to minimise it with respect to β.

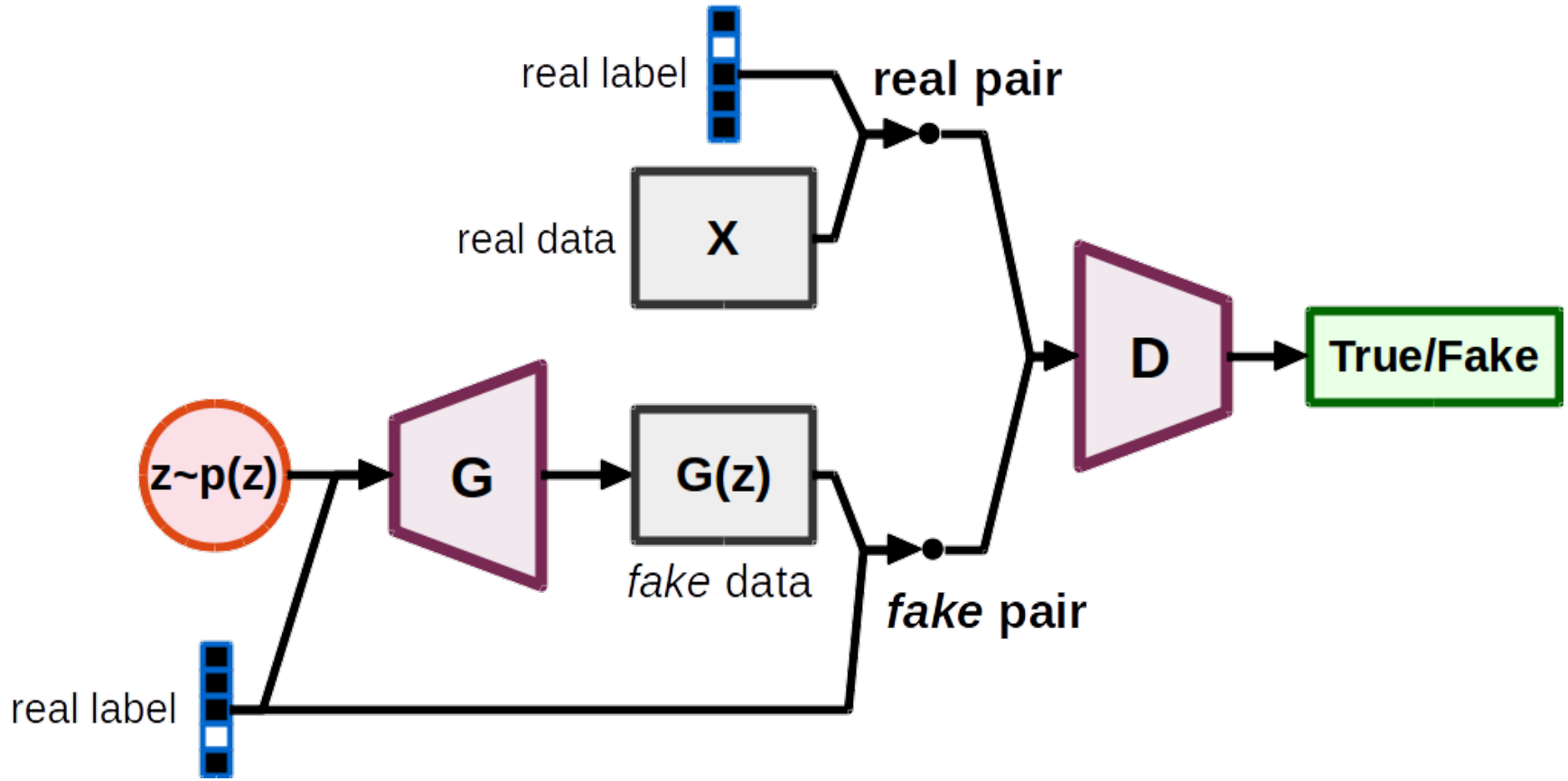
$$\min_\beta \max_\alpha \mathbb{E}_{x \sim p_{real}} [\log D_\alpha(x)] + \mathbb{E}_{z \sim p(z)} [\log 1 - D_\alpha\big(G_\beta(z)\big)]$$

- This is known as the adversarial (or GAN) loss as there are two parts trying to achieve two opposite goals.

Challenges:

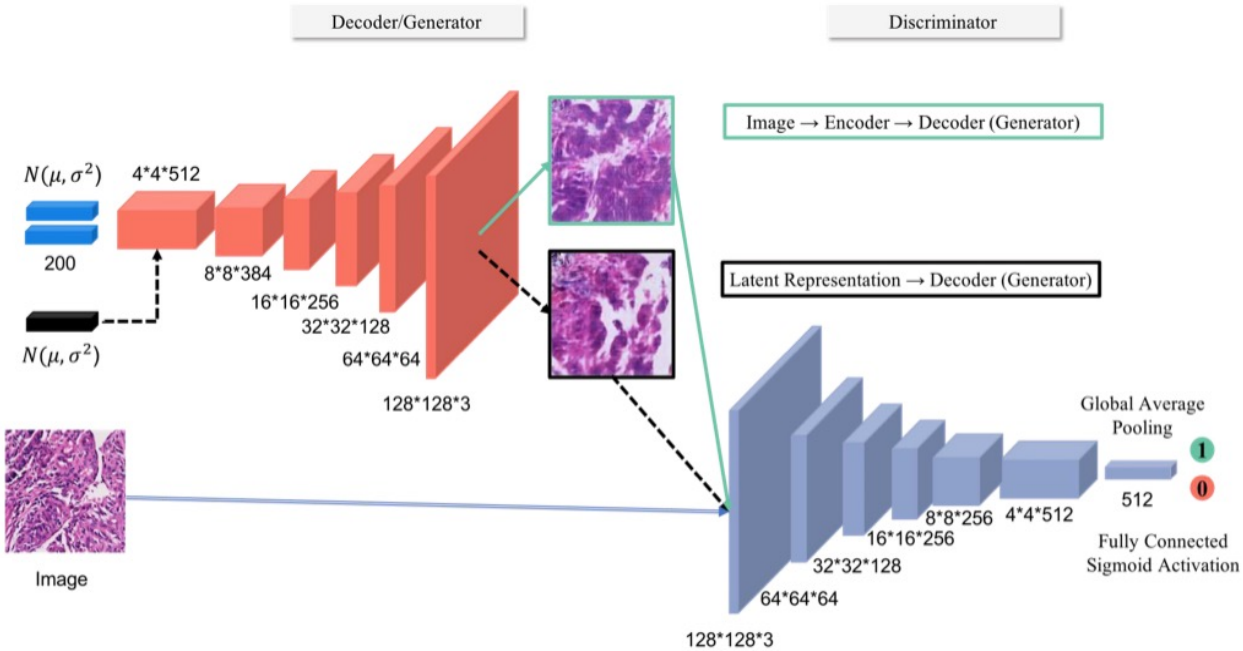
- **Unstable** optimization procedure
  - Generator and discriminator loss often oscillate without converging.
- Potential for **mode collapse**
  - Generator beats discriminator but only predicts same (or limited set of) image(s) every time
- Depends heavily on choice of hyperparameters
- Difficult to evaluate
  - When is an image realistic?

Conditional GAN (cGAN):

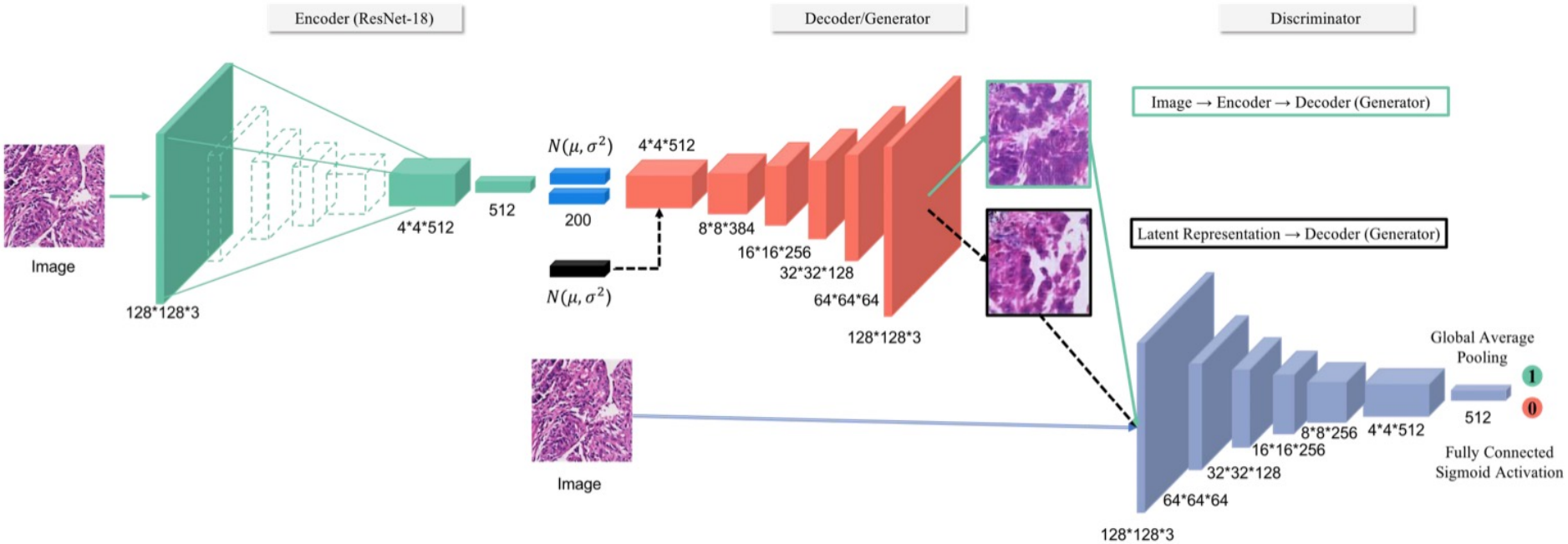


$D_\alpha(x|y)$ and $G_\beta(z|y)$ instead of $D_\alpha(x)$ and $G_\beta(z)$.

Hybrid models – VAE-GAN:

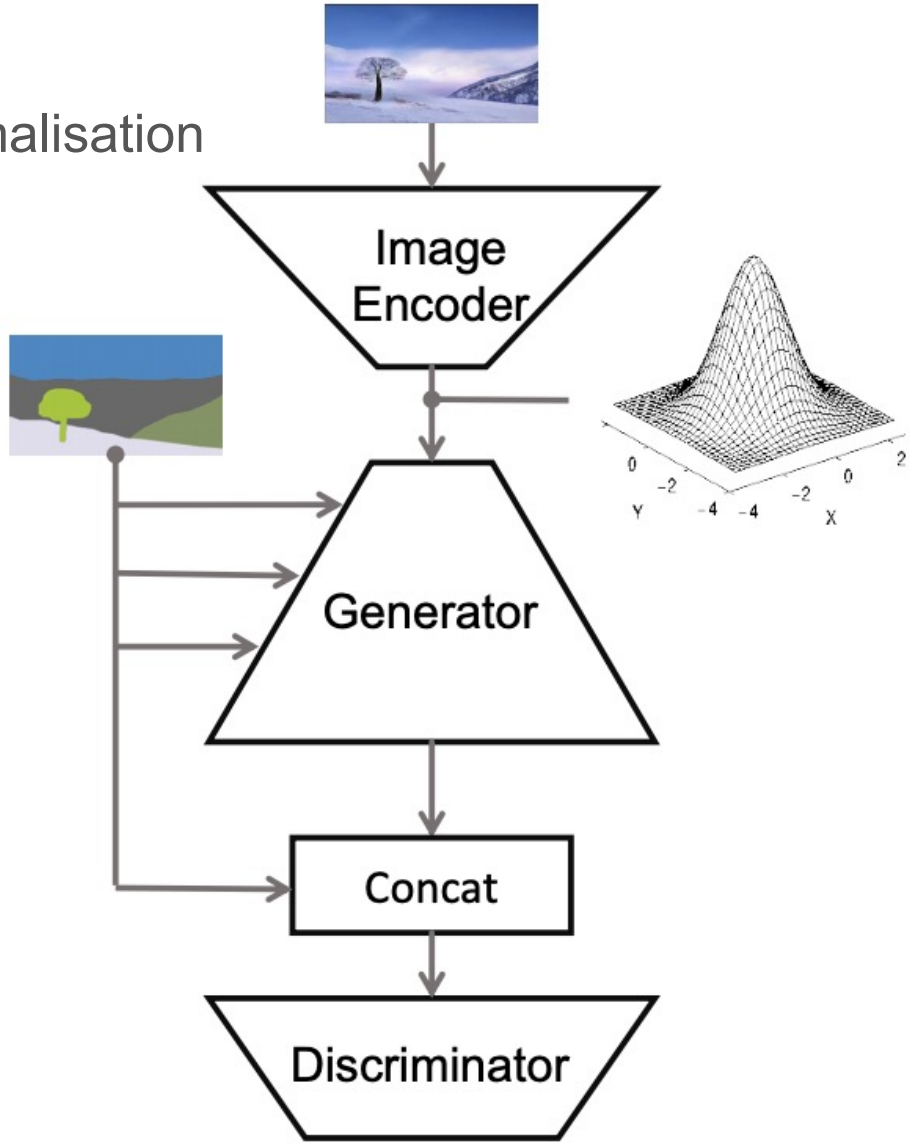


Hybrid models – VAE-GAN:

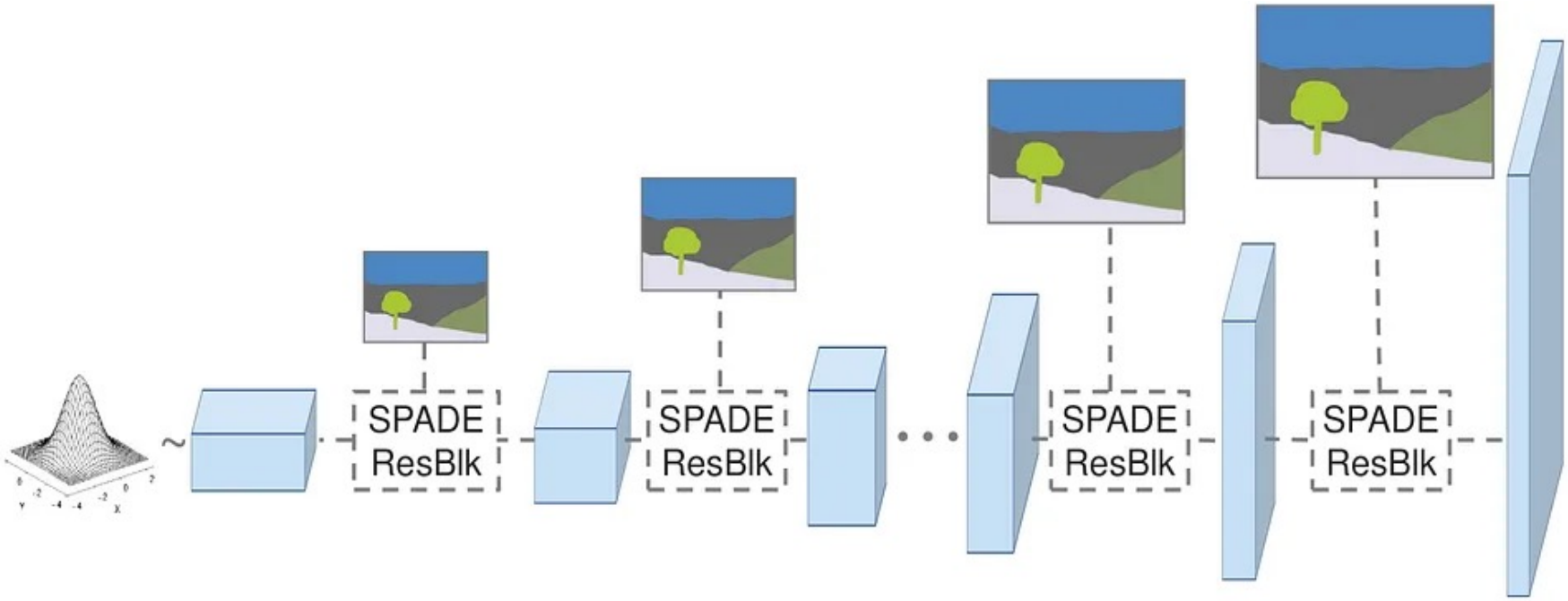


Condition on segmentations

Spatial adaptive normalisation (SPADE):



SPADE:



SPADE:

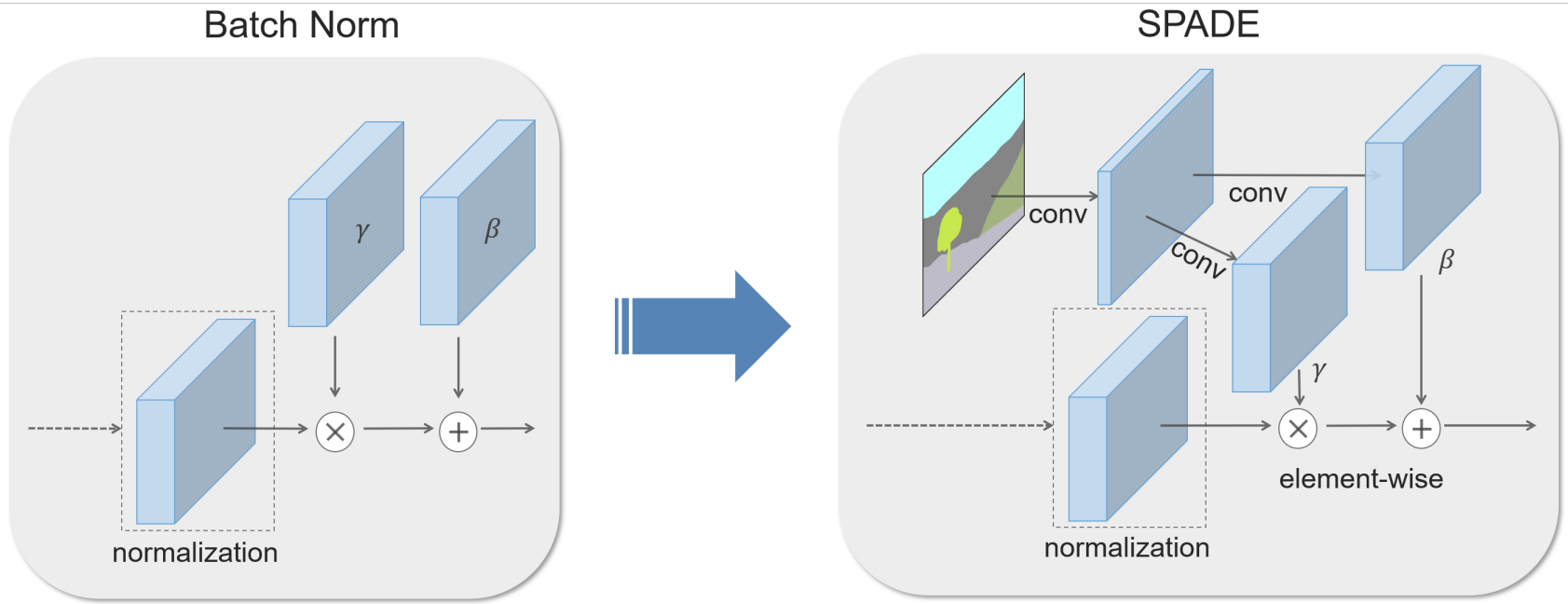
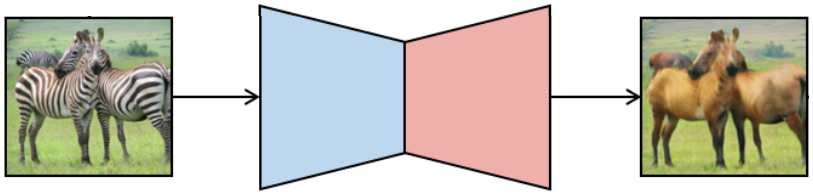
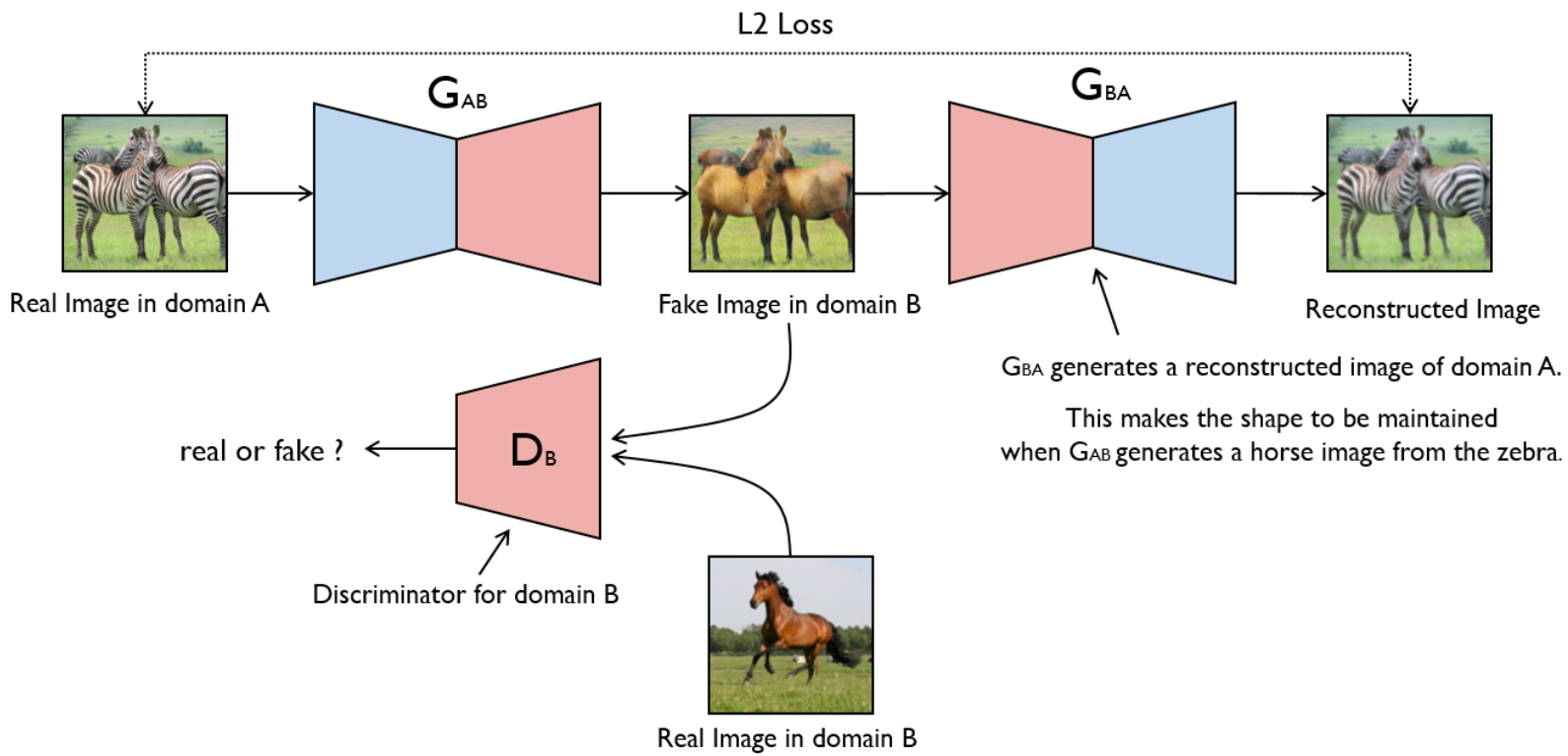

Batch Norm
γ
β
normalization
SPADE
conv
conv
conv
β
γ
element-wise
normalization

Image to image translation:



CycleGAN:

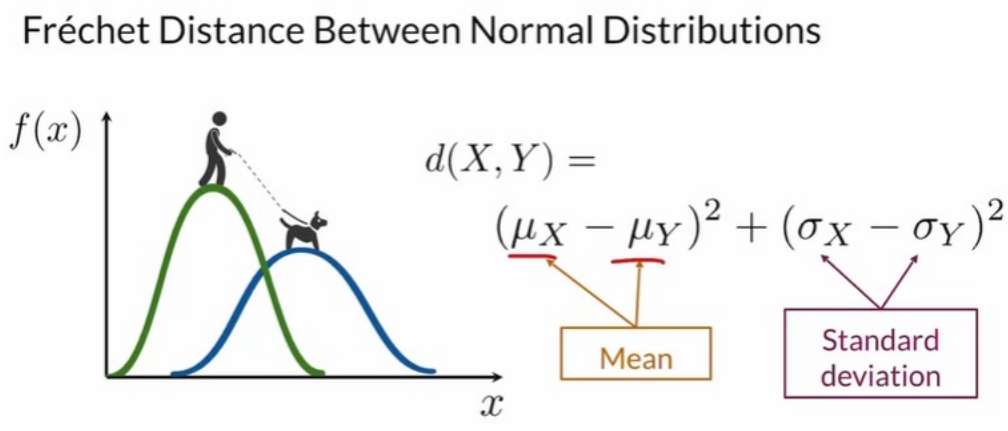

L2 Loss
G_{AB}
G_{BA}
Real Image in domain A
Fake Image in domain B
Reconstructed Image
G_{BA} generates a reconstructed image of domain A.
This makes the shape to be maintained
when G_{AB} generates a horse image from the zebra.
real or fake ?
D_B
Discriminator for domain B
Real Image in domain B

Evaluation

Evaluating synthetic data is very hard and still requires a lot of research!!

(especially for medical domains)

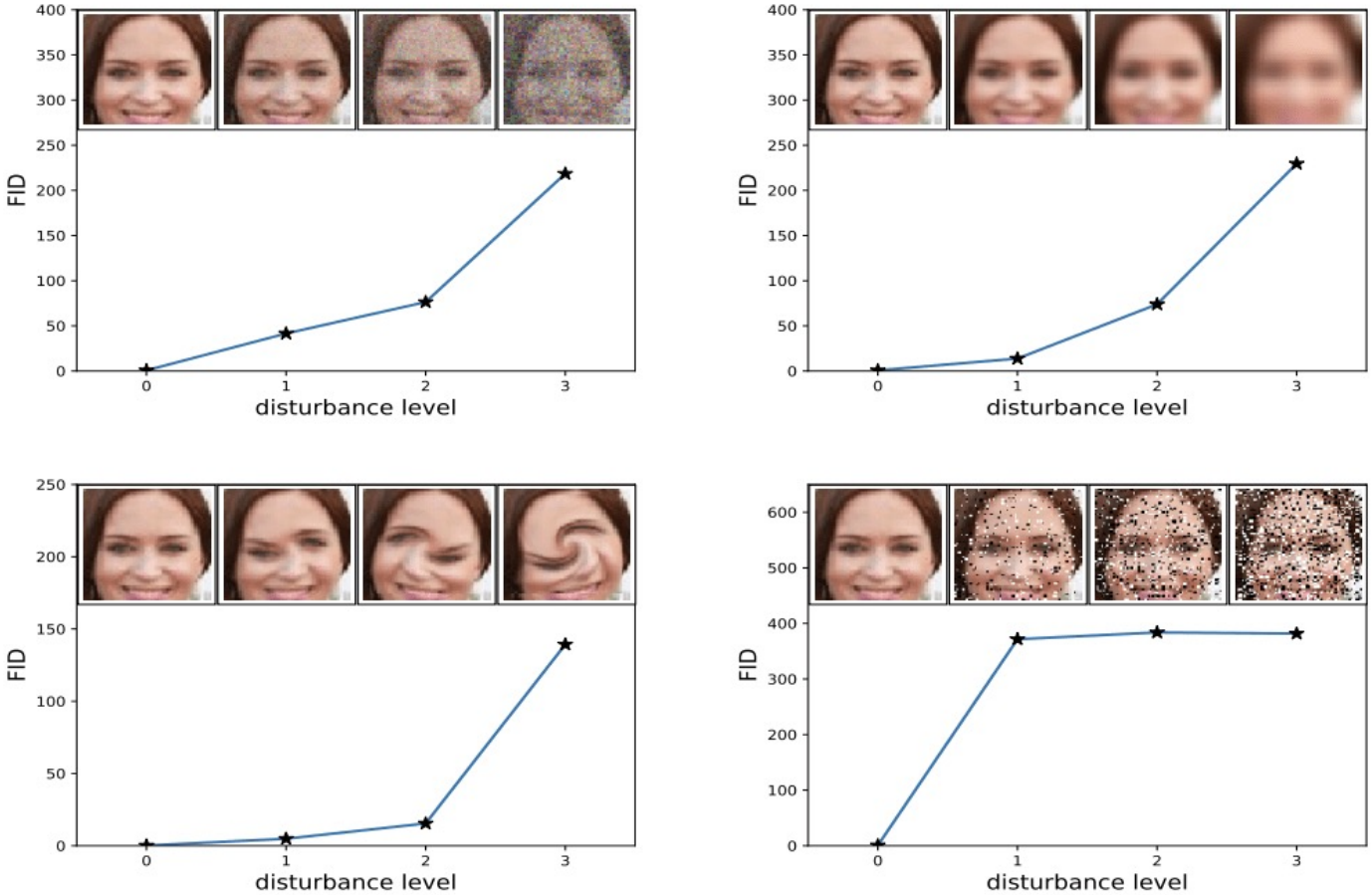
Fréchet inception distance (FID):



We use the activations from the pre-trained Inception V3 model to summarise each image and compute the distance between these features

$$FID = \|\mu_r - \mu_g\|^2 + T_r(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

FID example:

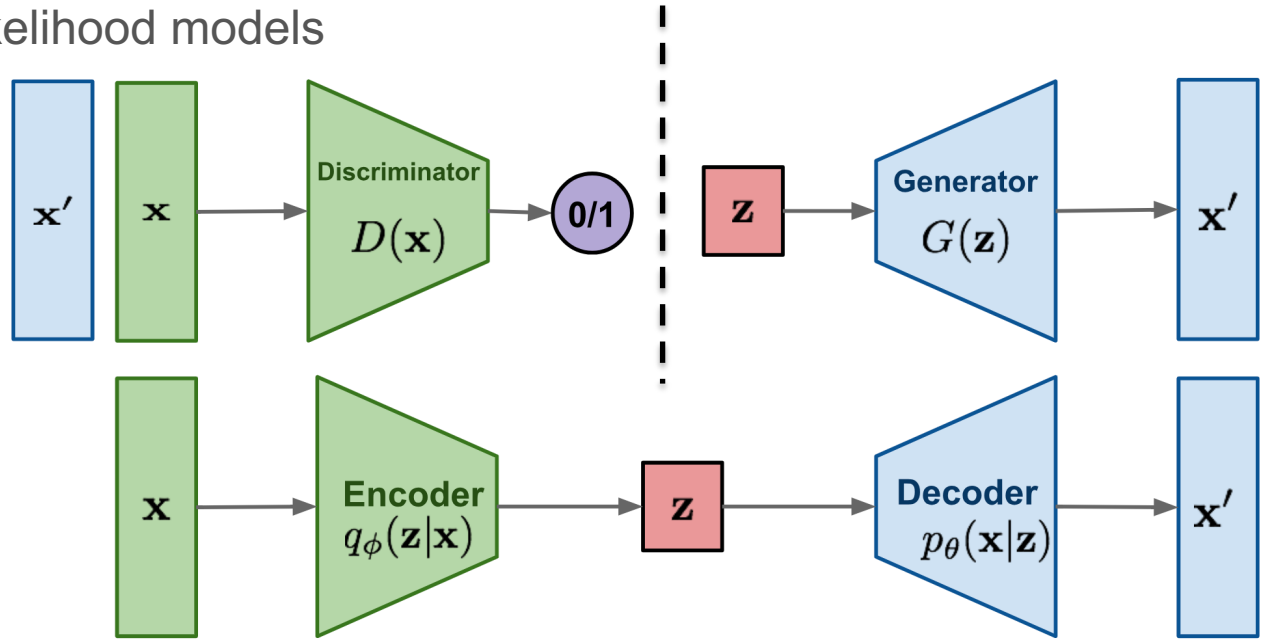

FID
disturbance level
0
1
2
3
0
50
100
150
200
250
300
350
400
500
600

Summary:

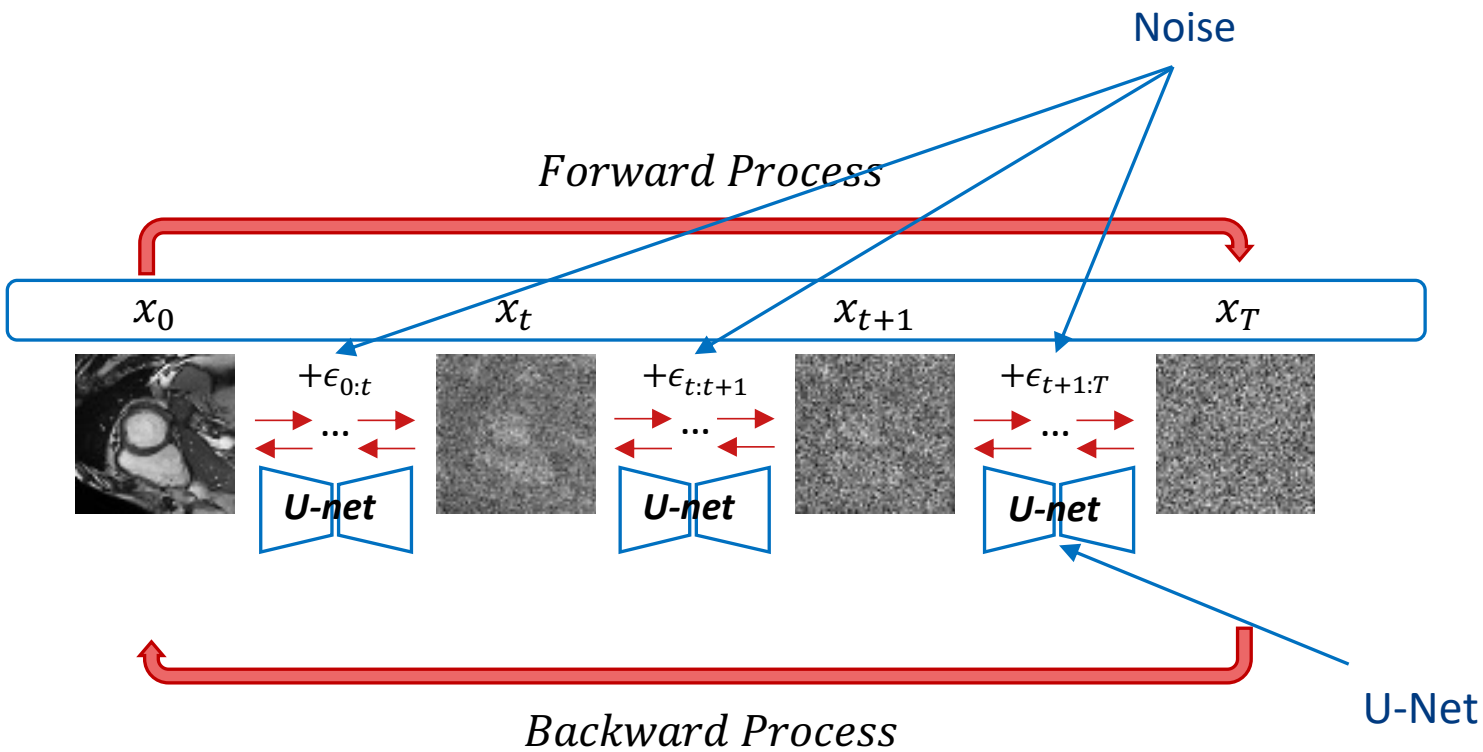
GANs are implicit likelihood models

GAN: minimax the classification error loss.

VAE: maximize ELBO.

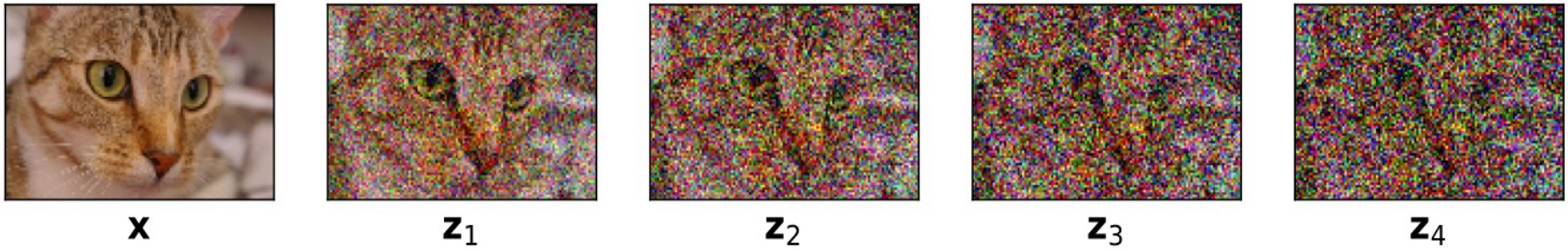


Denoising diffusion probabilistic model (DDPM):

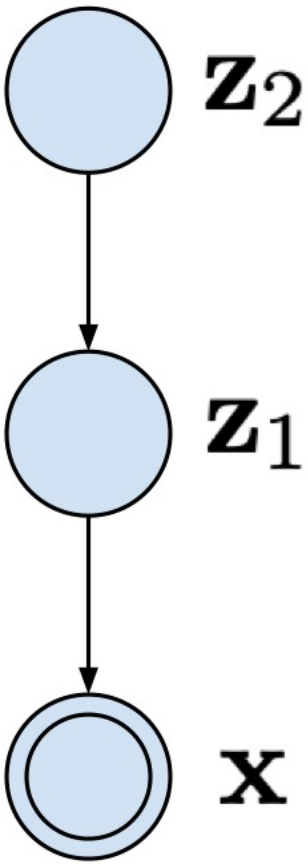


Diffusion models:

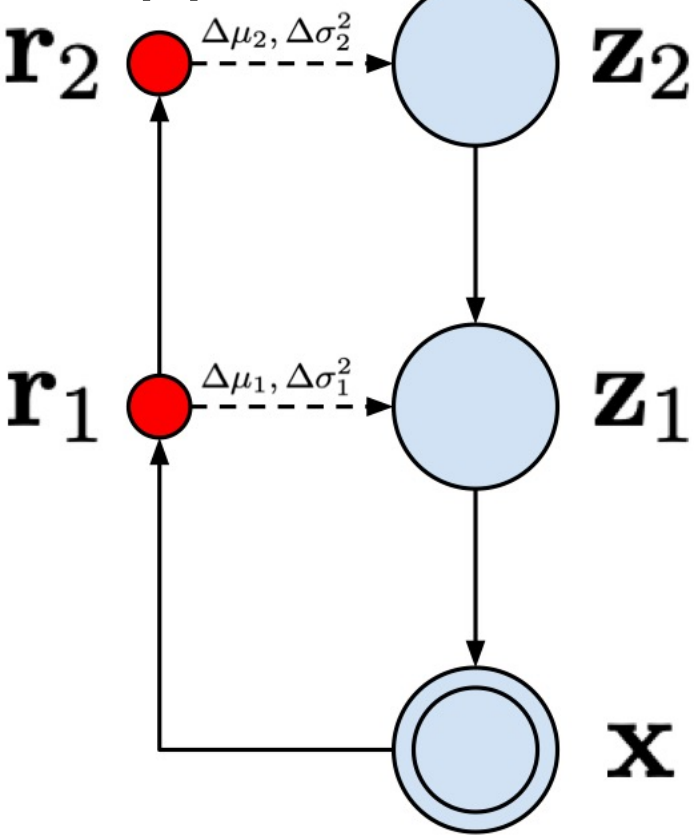
A diffusion model is a hierarchical VAE with the bottom-up path defined by a diffusion process and the top-down path parametrised by neural networks (reversed diffusion).



Hierarchical VAE:



Top-down hierarchical VAE with bottom-up path:



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Questions?